

(a) Nested Policy Engine

L_0	Token filter	per token	10^3 Hz	synchronous
L_1	Socratic guard	per response	1 Hz	synchronous
L_2	Tier refiner	per session	1.7×10^{-3} Hz	asynchronous
L_3	Memory layer	cross-session	1.2×10^{-5} Hz	asynchronous
L_4	Policy store	per revision	3.2×10^{-8} Hz	synchronous

Decisions compose by conjunction, so no layer can relax another.
Only the three synchronous layers gate a response.

(b) Monotone policy

t_5	ages 16–17	8/9 unblocked
t_4	ages 13–15	6/9 unblocked
t_3	ages 10–12	6/9 unblocked
t_2	ages 7–9	3/9 unblocked
t_1	ages 3–6	0/9 unblocked

Unblocked sets are nested; L_4 rejects any policy edit that breaks monotonicity of $a(c, t)$ in t .